



Software Defined Radios (SDR)

112-BOLCEW11

TLO – Terminal Learning Objective



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The student will identify the fundamentals of Software Defined Radios , in a classroom environment, given a networked computer workstation and appropriate references by demonstrating understanding on basic principles of a Software Defined Radio and characteristics of HackRF One Radio to support the application of EW effects critical to the unit's mission, in accordance with TC 9-64, achieving a minimum of 70% on all Practical Exercises (PE) and end of module exam.

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Learning Step Activities



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- Describe the Software Defined Radio basics
- Identify the principles of the Software Defined Radios
- Identify the characteristics of HackRF One Radio

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Software Defined Radio basics



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What is a Software Defined Radio



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Software-defined radio (SDR):

- Radio communication system where components that have been built into the hardware
- Produced by means of software on a personal computer or embedded system

Items implemented via software:

- Mixers
- Filters
- Amplifiers
- Modulators/demodulators

Technology is able to provide some significant advantages over traditional hardware based radios

- SDRs are being used in many different applications in many different areas

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Software Define Radio (SDR) Characteristics



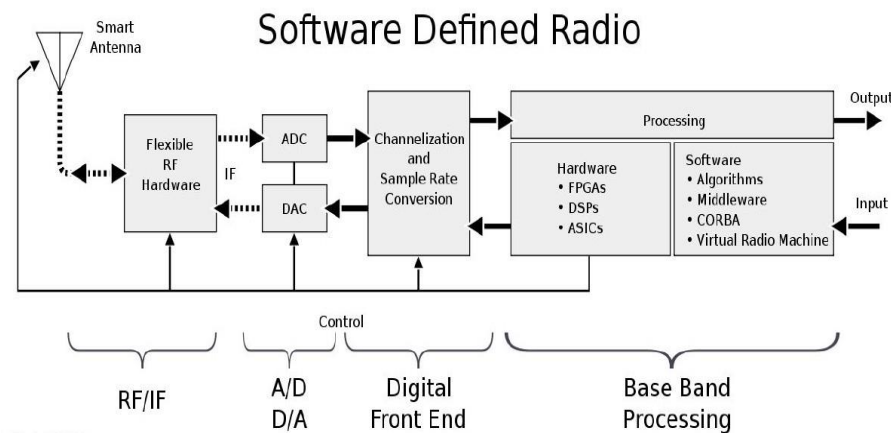
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Used to transmit and receive radio waves

- SDR emulating the radio hardware
- Computer application operates the controls

Refers to wireless communication in which the transmitter modulation is generated or defined by a computer.

The receiver uses a computer to recover the signal intelligence



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History of the “Digital Receiver”



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The term "digital receiver" was coined in 1970 by a researcher at a United States Department of Defense laboratory called the Gold Room.

- Created a software baseband analysis tool called Midas, which had its operation defined in software

The term "software radio" was coined in 1984 by in Garland, Texas by company called E-Systems Inc Division. (now Raytheon)

In 1984:

- Software Radio was a digital baseband receiver that provided programmable interference cancellation and demodulation
- Used for broadband signals

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Project “SpeakEasy” Phase 1: Software Defined Radio



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Software Defined Radios were established in 1995 by Stephen Blust.

- Concept was first implemented with an Intermediate Frequency (IF) Analog to Digital Converter (ADC) (IF ADC) in the early 1990s

Software-defined radios have their origins in the U.S. and European defense industry.

- Air Force led military project named SpeakEasy (phase 1) (1990-1995)

The primary goal of the SpeakEasy project was to :

- Demonstrate a radio tactical ground air control party that could operate from 2 MHz to 2 GHz
- Interoperate with ground force radios

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SpeakEasy Phase II (1 of 2)



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SPEAKeasy phase II goals were:

- Quickly reconfigurable architecture
- Hold several conversations at once, in an open software architecture
- Cross-channel connectivity
- Make it smaller, cheaper, and weigh less

Project was so successful that the demonstration was halted and was pushed into production early.

- Radio went into production with only a 4 MHz to 400 MHz range

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Speakeasy Phase II (2 of 2)



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Goals of Speakeasy PH II:

The software architecture identified standard interfaces for different modules radio:

- Radio frequency control - to manage the analog parts of the radio
- Modem control - managed resources for modulation and demodulation schemes (FM, AM, Single-sideband modulation (SSB), and Quadrature amplitude modulation (QAM).

Waveform processing module - performed the modem functions

- "key processing" and "cryptographic processing" managed the cryptographic functions

Multimedia module - performed voice processing "a human interface"

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SDR Concept (1 of 2)



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The ideal receiver scheme would be:

- To attach an analog-to-digital converter to an antenna
- A digital signal processor would read the converter
- Its software would transform the stream of data from the converter to an application

An ideal transmitter would be similar:

- A digital signal processor would generate a stream of numbers
- These would be sent to a digital-to-analog converter connected to a radio antenna

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SDR Concept (2 of 2)



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Most receivers use:

- Variable-frequency oscillator
- Mixer
- Filter

To tune the desired signal to a common intermediate frequency or baseband, It must be sampled by the analog-to-digital converter after amplification.

- An intermediate frequency (IF) is a frequency to which a carrier wave is shifted as an intermediate step in transmission or reception

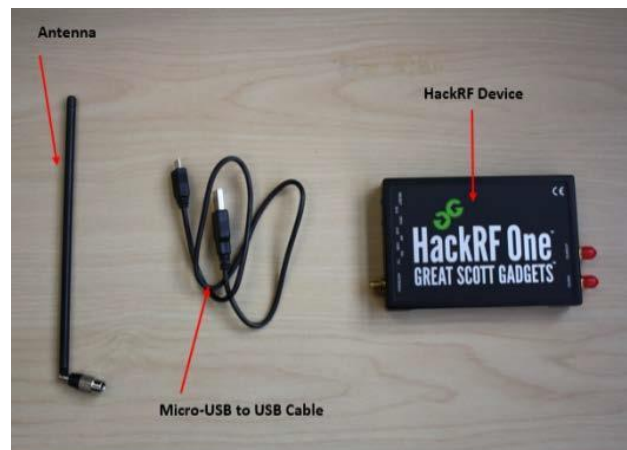
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HackRF One SDR

SDR components:

- Computer
- Radio device
- HackRF One
 - The HackRF One device
 - Micro- USB to USB cable
 - ANT500 antenna



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Baseband (Frequency)



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Baseband Frequency:

- RF signal that has a near-zero frequency range

Baseband signals are transmitted without modulation:

- No shift in the range of frequencies of the signal
- Low-frequency contained within the bandwidth frequency
 - Close to 0 hertz up to higher cut-off frequency

Baseband can be synonymous with lowpass or non-modulated

Baseband is different from:

- Passband, bandpass, carrier-modulated, intermediate frequency, or radio frequency (RF)

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Radio Evolving (1 of 2)



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Rapidly evolving capabilities of digital electronics:

- Enables numerous practical processes for SDRs which were once only theoretically possible

Basic SDR system consists of:

- Personal computer equipped with a sound card
- Preceded by some form of RF front end (i.e. HackRF one)
- Significant amounts of signal processing are handed over to the general-purpose processor
- Design gives a radio that can receive and transmit many radio protocols or waveforms

Long term:

- SDR are expected to become the dominant technology in radio communications

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Radio Evolving (2 of 2)



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SDRs have significant utility for the military and cell phone services:

- Serves a wide variety of changing radio protocols in real time
- Flexible to avoid "limited spectrum" from previous kinds of radios

SDRs uses Spread spectrum and ultra-wideband (UWB) techniques:

- Allow several transmitters to transmit in the same place on the same frequency
- Little interference
- UWB Technique: communication method used in wireless networking that uses low power to achieve high bandwidth connections

Software defined antennas adaptively "lock onto" a directional signal

- Receivers can better reject interference from other directions
- Allows to detect fainter transmissions

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Joint Tactical Radio System (JTRS)

- US military to produce radios that provide flexible and interoperable communications

Examples of radio terminals:

- Hand-held radios
- Vehicular radios
- Airborne radios
- Dismounted radios
- Base-stations radios

Goal is achieved through the use of SDR systems

- Internationally endorsed open Software Communications Architecture (SCA)
- Program provides flexible approach meeting Army communications needs through software programmable radio technology

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Amateur software radio uses a direct conversion receiver

- Mixer technologies used are based on the quadrature sampling detector and the quadrature sampling exciter

SDR software performs all:

- Demodulation
- Filtering (both radio frequency and audio frequency)
- Signal enhancement (equalization and binaural presentation)

GNU Radio uses Universal Software Radio Peripheral (USRP)

- USB 2.0 interface
- FPGA
- High-speed set of analog-to-digital and digital-to-analog converters, combined with reconfigurable free software

Advantages



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Software radio has two major advantages:

- Flexibility
 - Being able to switch channels, change modulation has always been valuable to the military
 - SDR can be programmed for emerging standards
- Ease of adaptation:
 - Virtually any aspect of a program that implements radio functions can be easily changed

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Terms used in SDR (1 of 4)



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Radio - Technology for wirelessly transmitting or receiving electromagnetic radiation to facilitate transfer of information

Radio Node - A radio point of presence incorporating a radio transmitter or receiver

Software - Modifiable instructions executed by a programmable processing device

Physical Layer - Layer within wireless protocol which processing of RF, IF, or Baseband signals including channel coding occurs

Digital Signal Processing (DSP)

- Signals processed via analysis, modification, and synthesis by a sequence of numbers
- Represent samples of a continuous (e.g. time, frequency)

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Terms used in SDR (2 of 4)



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Data Link Layer

- Protocol responsible for reliable frame transmission over a wireless link
- Through the employment of proper error detection and control procedures

Software Controlled - Use of software processing within the radio system or device to select the parameters of operation.

Software Defined - Use of software processing within the radio system or device to implement operating (but not control) functions.

Software Controlled Radio - Radio in which some or all of the physical layer functions are software controlled.

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Terms used in SDR (3 of 4)



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Transmitter - Apparatus producing radio frequency energy for the purpose of radio communication.

Receiver - Device that accepts a radio signal and delivers information extracted.

Air Interface

- Subset of waveform functions designed to establish communication between two radio terminals
- Waveform equivalent of wireless physical layer and the wireless data link layer

Waveform

- Set of transformations applied to information to be transmitted
- Corresponding set of transformations to convert received signals back to their information content or digital

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Terms used in SDR (4 of 4)



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GNU Radio Companion - Graphic user interface (GUI) software; allows users to create flow graph that visualizes and processes signals

Filter - Clean up received signals in order to limit unnecessary noise and interference

- Clean up signals when transmitting to cause less radio interference

Hertz (Hz) - Derived unit of frequency

Receive - The act of obtaining and interpreting radio waves from other devices (radio Receive)

Sample Rate - Rate at which data is taken digitally over a period of time. Measured in Hz

Transmit - The act of sending out a signal via a wire or radio waves

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Questions?

Principles of Software Defined Radios



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Why Your Next Radio Will Be SDR



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What is the Nyquist Theorem?



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Nyquist Theorem is a principle that engineers follow in the digitization of analog signals.

- Converts analog to digital (ADC)
- Takes what is called SAMPLES
- Sampling rate is the number of samples per second

The sampling rate must be at least $2f_{\max}$, or twice the highest analog frequency component.

- Actuated by a pulse generator (clock)
- Digital signal is converted back to analog form by a digital-to-analog converter, false frequency components appear
 - Undesirable condition is a form of distortion called aliasing

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Nyquist Theorem in Practice



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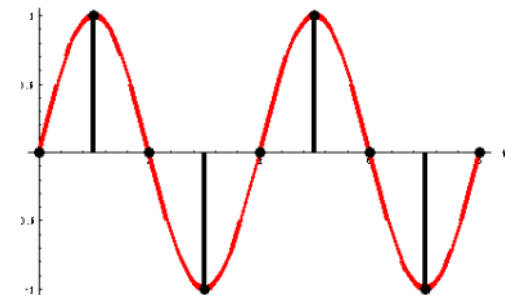
Nyquist Theorem – To adequately reproduce a signal it should be periodically sampled at a rate that is 2X the highest frequency you wish to record.

To get images into a form, computers can manipulate. They must be modified from their analog form into a digital form, this is done by sampling.

Sampling - A technique used to record analog information by recording periodic snapshots

- If the sampling rate is fast enough, the eye cannot discern the gaps between each snapshot when they are played back
- This is the principle behind motion pictures.

Sampling is the key technique used to digitize analog information such as sound, photographs, and images



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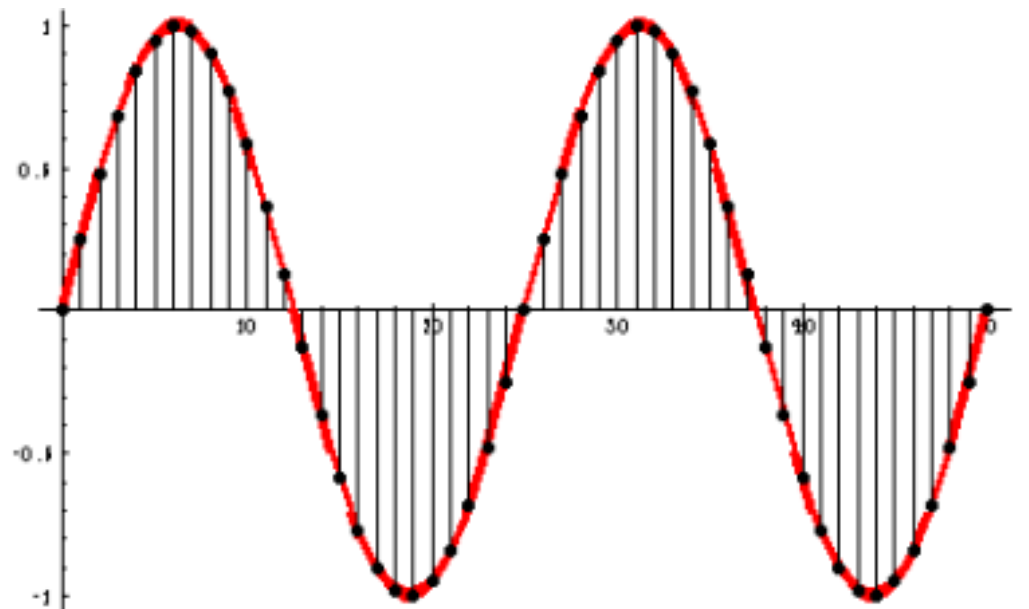
Oversampling



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A High Sampling Rate equals a much greater than 2 times the highest frequency.

This is Oversampling, while not bad it will take up more time, create a large digital file and use up hard drive space on your computer.



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Under sample

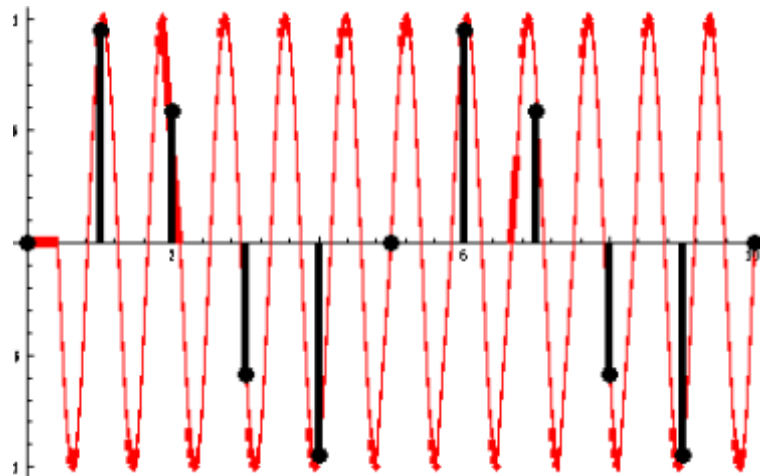


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Under sampled - low sampling rate produces results that report false information about the analog data; which does not represent the original.

This phenomenon is called aliasing.

Aliasing does not occur if the sampling rate is greater than twice the frequency of a desired information.



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IQ Data

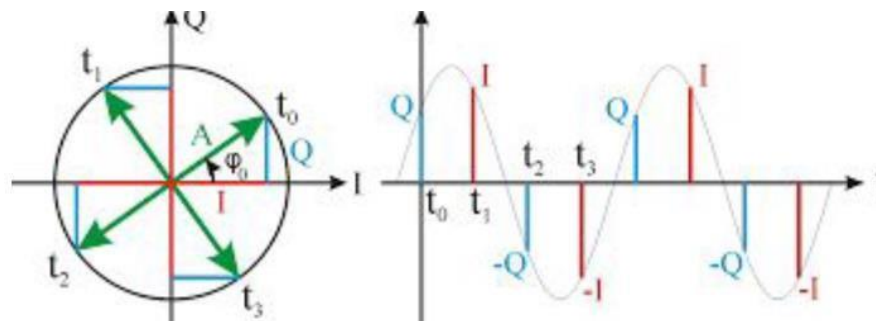


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In the RF world, it is common practice to sample complex values.

Samples are sequence of elements with 32 bit float values, one for the real component (I, in-phase) and one for the imaginary component (Q, quadrature).

- IQ sampling doubles your sampling bandwidth.
- Complex sampling does not "break" Nyquist.
- IQ quadrature sampling produces twice as many bits per second of information.



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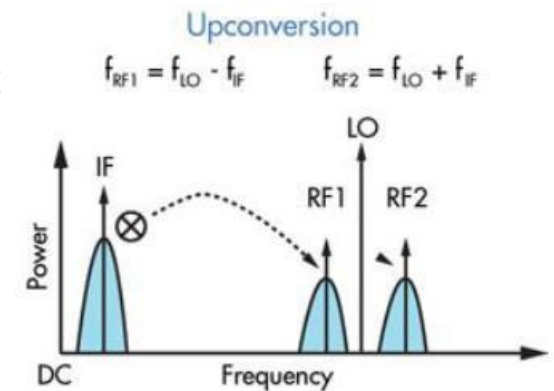
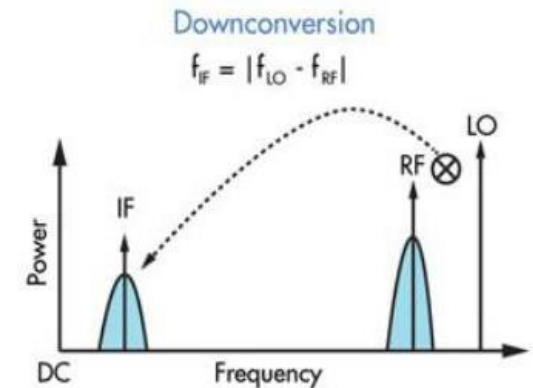
RF mixing



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How are RFs captured above the Nyquist rate for that ADC?

- RF mixing
- A frequency mixer that creates new frequencies from two signals applied to it
- Most common application: two signals are applied to a mixer and it produces new signals at the sum and difference of the original frequencies



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Analog Filtering



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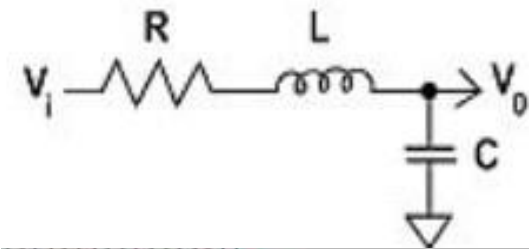
Analog filters:

- Removes signals above a certain frequency
- Called low pass filters
- Allows low frequency signals pass through the filter while blocking everything above the cutoff frequency

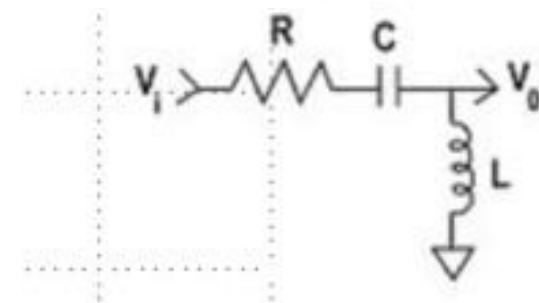
High Pass: Filter that removes all signals below a certain frequency

- It lets pass everything higher than the cutoff frequency

(A) LOW PASS



(B) HIGH PASS



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Digital Filtering



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Contains a ADC which samples the signal as a input:

- Micro-processor does the work
- Software based

Two classes of digital filters:

- Finite Impulse Response (FIR) : filter whose impulse response (or response to any finite length input) is of limited duration
- Infinite Impulse Response (IIR): Distinguished by having an impulse response which does not become exactly zero past a certain point, but continues indefinitely

Advantage of IIR vs FIR:

- Lower order or number of terms can be used
- Less calculations needed

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Radio Frequency Filtering



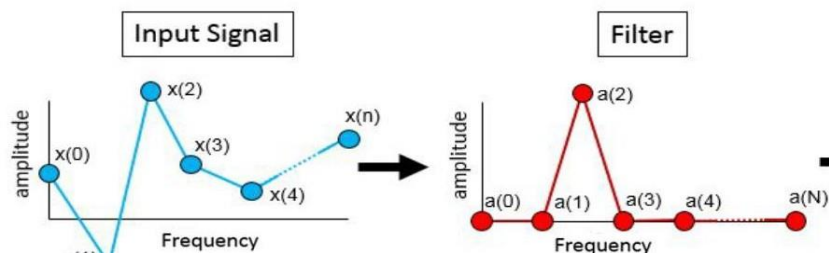
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Radio Frequency (RF) Filters:

- Special circuits that allow a signal of interest to pass through while attenuating any undesired signals
- Undesired signals add interference and noise

There are four types of filters:

- Low pass filter
- High pass filter
- Band pass filter
- Band reject filter



Each different type rejects or accepts signals in a different way.

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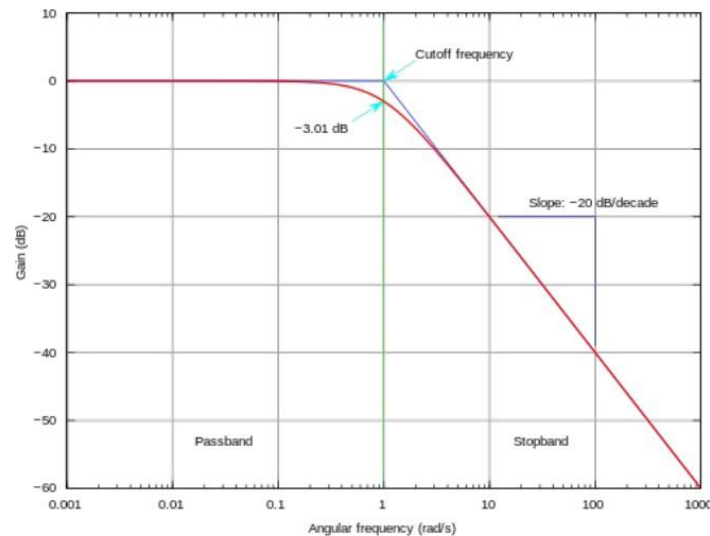
Low Pass Filter



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Low-pass filter (LPF):

- Filter that passes signals with a frequency lower than a selected cutoff frequency
- Attenuates signals with frequencies higher than the cutoff frequency
- Typically it is nominally flat until the cut-off point, and then it rolls off



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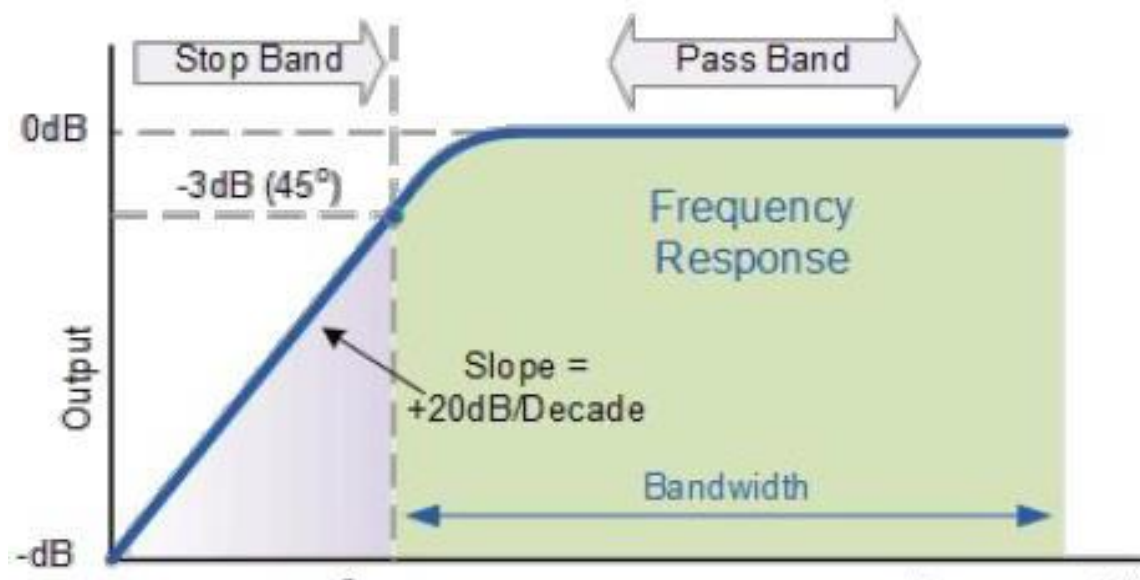
High Pass Filter



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High-pass filter (HPF):

- Electronic filter that passes signals with a frequency higher than a certain cutoff frequency
- Attenuates signals with frequencies lower than the cutoff frequency



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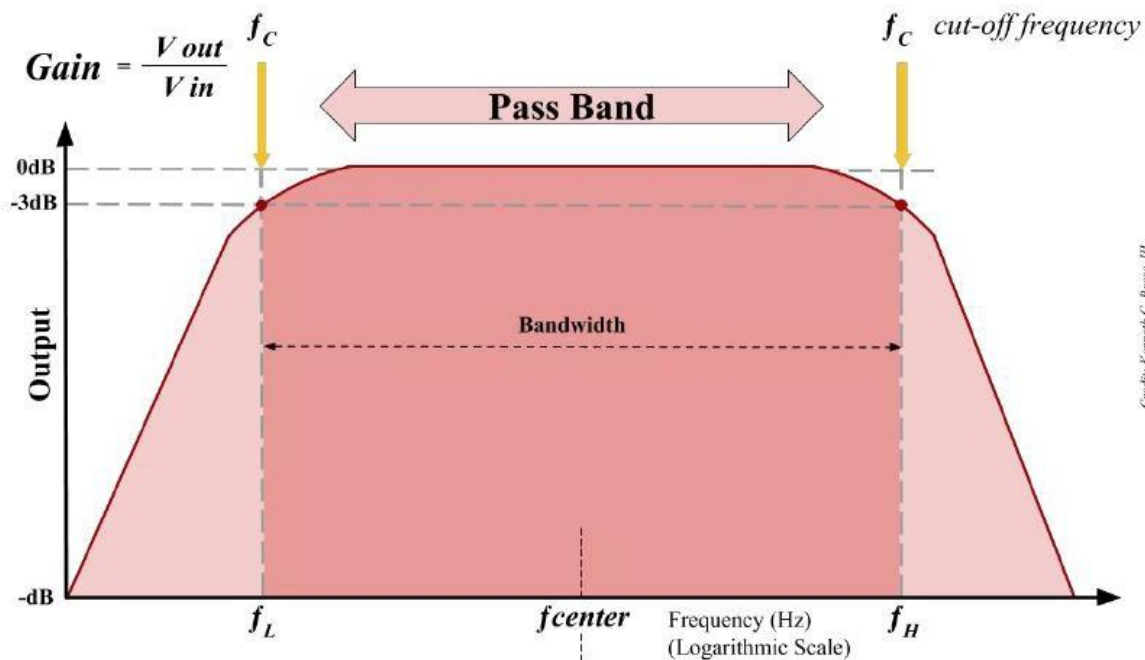
Band-Pass filter



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Band-Pass Filter:

- Device that passes frequencies within a certain range
- Rejects (attenuates) frequencies outside that range



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Re-sampling



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Resampling (also known as sample rate conversion):

- General term that refers to changing the sample rate of a digital representation of a signal while preserving, as closely as possible, the information contained in the original signal
- Based on two signal processing operations: Decimation and Interpolation

Two types: decimation and interpolation:

- Decimation is removing samples from a sampled signal to reduce the sampling rate
 - Interpolation is adding samples to increase the sampling rate
- In both cases information about the signal is reduced.

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Decimation



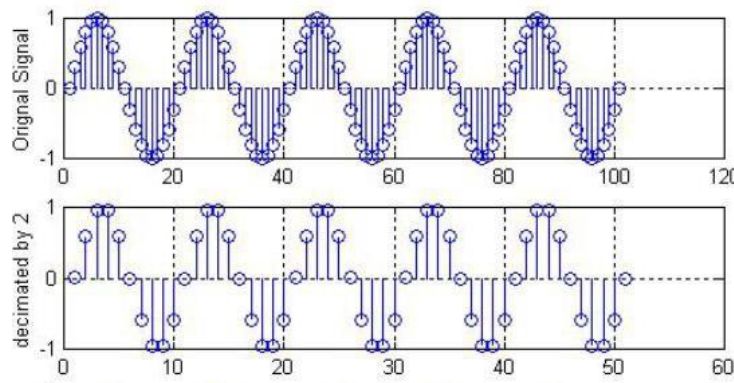
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Decimation:

- Process of reducing the sampling rate
- Removing samples saves space
- Uses low-pass filtering of a signal, then throws away the unused samples

Example: FM radio has a 15 kHz wide audio band

- Sample in the MHz range for initial signal capture
- Could resample signal down to 30 kHz to save time and space



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Interpolation



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Interpolation is increasing the sampling rate of a given signal (up sampling), increases the spectral separation between the images of the original spectrum (I phase and Q phase).

Two types Interpolation:

- Padded adds “blank” zero samples between signal samples
- Approximately estimates the value of the sample to be added

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Demodulation



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What is Demodulation?

- Extracting original information bearing signal from a carrier wave
- Systems use what is called a “Demodulator” to perform the process

Demodulator:

- Electronic circuit (or computer program in a software-defined radio) that is used to recover the information content from the modulated carrier wave

SDR types of modulation for the lesson:

- Amplitude modulation (AM)
- Frequency modulation (FM)
- Phase modulation (PM)

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AM Demodulation



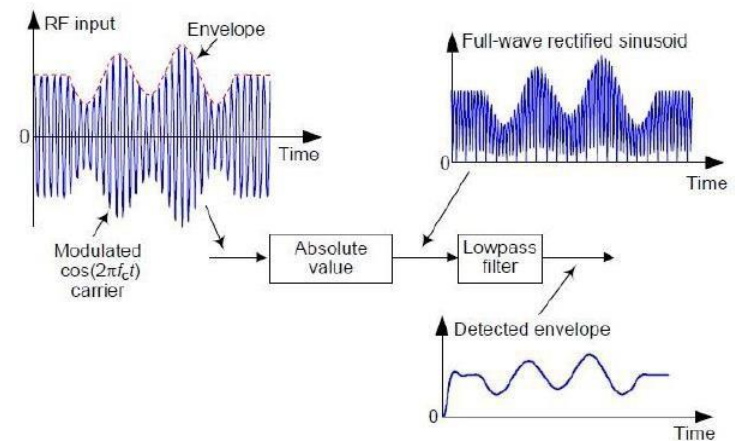
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AM Demodulation:

The simplest form of AM demodulation is envelope detection.

Demodulates a previously modulated signal by removing all high frequency components of the signal.

Simply take the absolute value (or magnitude) of the signal to do full wave envelope detection.



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FM Demodulation

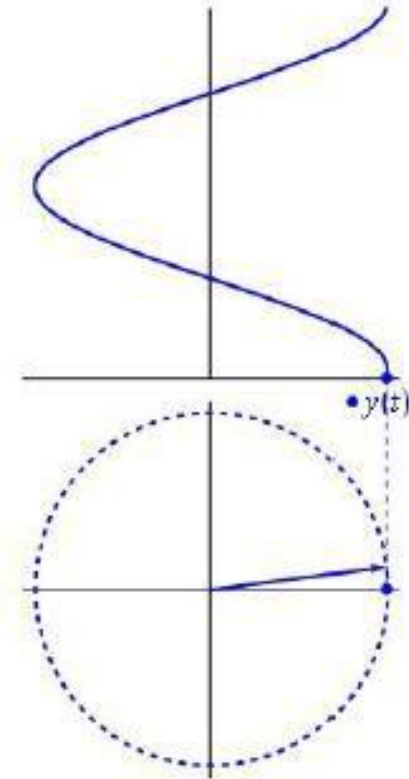


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FM demodulation:

- Known as Quadrature Demodulation
- Produces two baseband waveforms that, when taken together, convey the information that was encoded into the carrier of the received signal

Take the angle between two sample points that shows the instantaneous frequency change with respect to time



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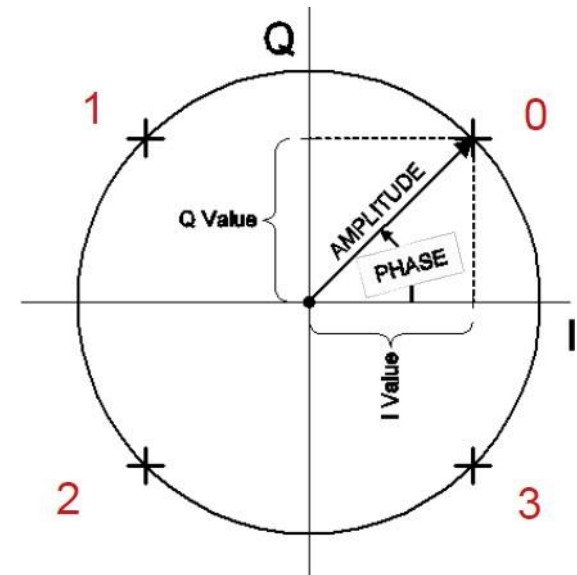
PM Modulation



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PM demodulation in SDR:

- Separating I and Q in this way allows the ability to measure the relative phase of the components
- Two copies of the local oscillator are generated
- One 90 degrees delayed with respect to the other
- These are separately mixed with I and Q
- Preserves the phase relationships through the conversion



Example:

QPSK demodulation only cares about the sign of the I and Q data (in order to determine quadrant)

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Questions?

Characteristics of HackRF One Radio



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Background



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The HackRF One is a Software Defined Radio (SDR)

- Device with the ability to digitize radio signals that are received or transmitted by the device
- Works with frequencies from 1MHz to 6 GHz, which is the same frequency range as:
 - Bluetooth
 - FM radio
 - Cellular technology



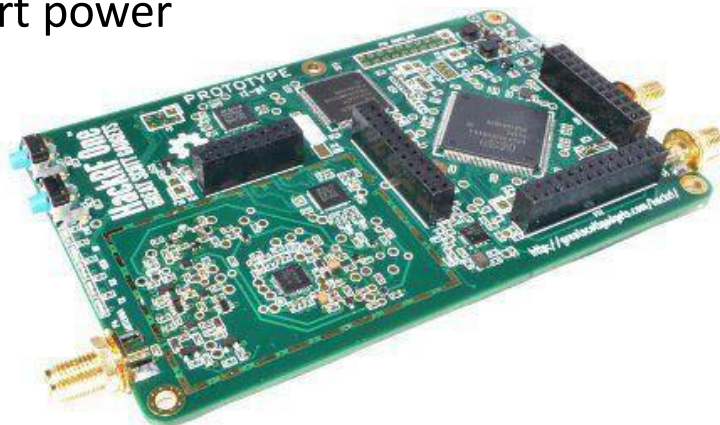
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HackRF One Characteristics



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- Frequency Range: 1 MHz to 6 GHz:
- ADC Resolution: 8 Bits
- Max Bandwidth: 20 MHz
- TX/RX: TX and RX (Half Duplex)
- Supported Sample rates: 2Msps to 20Msps (QUADRATURE)
- Software controlled antenna port power
- Portable
- OPEN SOURCE



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Physical Characteristics



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Reset Button - Used to reboot the HackRF One

3v3 LED, 1V8 LED, RF LED - All three of these LEDs are used to indicate power

USB LED - Indicates that the HackRF One is communicating over USB

DFU Button - Used to install or update the firmware if it is not working properly

RX LED - An orange light that indicates that the device is receiving information

TX LED - A red light that indicates that the device is transmitting information



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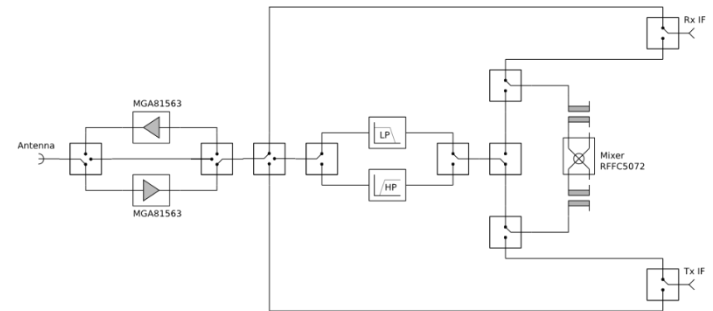
HackRF Data Flow



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HackRF has a very high flexibility when choosing the signal path:

- Radio Frequency (RF) switches are located at important points to allow the selection of various components (based on user programming)
- After the antenna input two amplifiers follow:
 - One serves the input
 - The other the output path.
- The amplifier block is followed by:
 - Low-pass and a high-pass filter
 - Can be used to limit the input/output signal
- After the filter the signal arrives at an RF mixer, the signal is mixed up or down. IF signals to be switched directly to the amplifiers or directly to the antenna



HackRF One Frontend Block diagram
based on frontend schema dated 13. Feb. 2014
(C) Eki Plicht, DF4QR WIMo

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Questions?

Summary



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In summary we discussed:

- Software Defined Radios Basics
- The Principle's of Software Defined Radios
- The Characteristics of HackRF One Radio

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